

Software Requirements Report for Software Engineering Group Project

2023/2024 Semester 2
<Fitness Trainer Appointment System>
<April 1, 2024>

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Introduction

This project aims to develop a web-based Trainer Appointment System for a fitness center, built on the *SpringBoot* framework and *Thymeleaf* engine. The system allows members to make appointments according to course, time, and specific trainer, while trainers and managers can use their own portals to conduct corresponding operations. In the design process, we categorized requirements into three priority levels:

1. Important requirements: These are core functionalities of the appointment system. The entire system cannot function properly without these functions, including member account management, booking process etc.
2. Normal requirements: These are not essential for the core system but can provide basic service for users. For example, reset and finding password, viewing and editing personal information, etc.
3. Insignificant requirements: These features mainly aim to enhance user experience which make the system more comprehensive, while their absence won't affect the system's core function. For example, class time reminders, announcement posting, etc.

The development will adapt a hierarchical approach based on the priority level of requirements to ensure the efficiency of the development and reliability of the software.

I was mainly responsible for three PBIs related to course management and two PBIs related to trainer tip-up at different user terminals. The contents related to course management are very important because they are the core part of the Fitness Trainer Appointment System, especially the “Add and modified course information”, which will also be prioritized in future development work. The two PBI of trainer tip-up are important supplementary functions for physical gyms. These two PBIs will be developed gradually after the core functions are developed.

Product Backlog Item

PBI #	TAS-Man-007
Title	Add and modified course information

Creation Date	April 1, 2024	Status	New	Sprint	
Priority	Important	Effort		Business Value	

Story

As the manager, I want to add new courses so that member users (logging in) to the website can see all the available courses on the website. At the same time, I can also modify it in time to ensure the timely update of information.

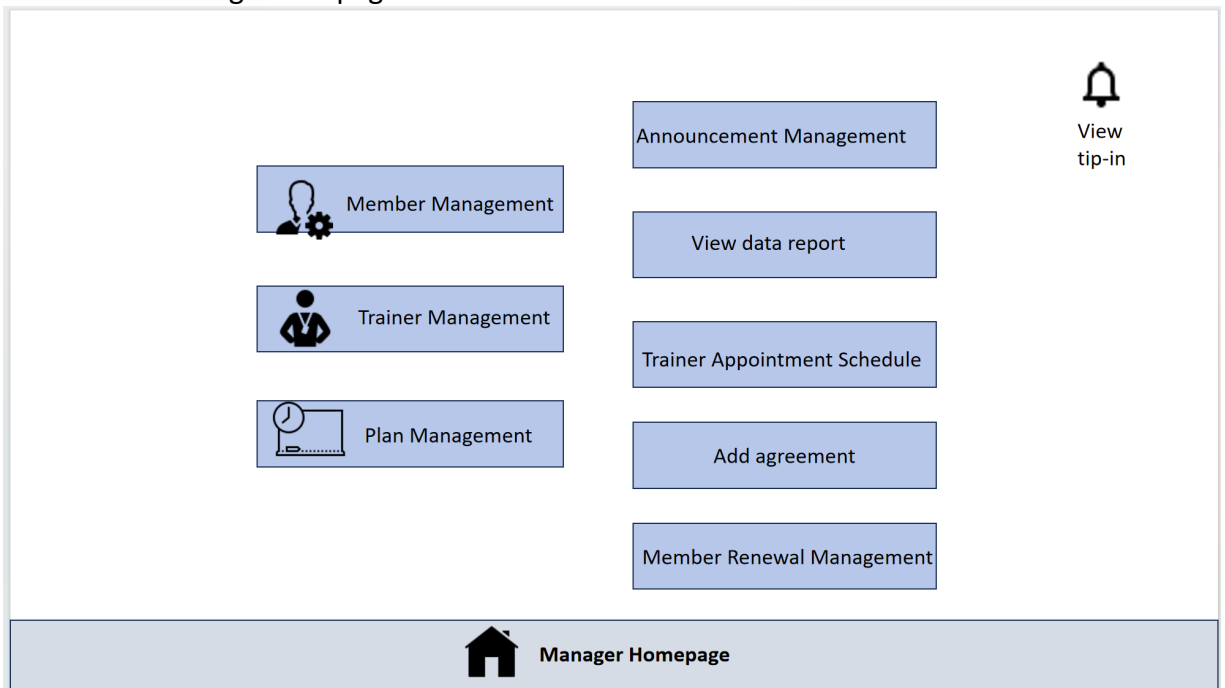
Description

1. Each course should have a regular instructor.
2. Each course should have a fixed time.
3. Each course should have a fixed location.
4. Each course should have a limited or unlimited number of people.
5. If the corresponding time of the course trainer conflicts or the venue conflicts at the same time, it should not be added successfully.
6. The duration of the course can be modified.
7. The instructor of the course can be modified.
8. The location of the course can be modified.
9. Inaccurate or conflicting changes should be rejected with an error.

Acceptance Criteria

- 1) Given that I am a valid manager, when I navigate to the "Add Course" section on the "Plan Management" page, then the "Add Course Information" page will be displayed.
- 2) Given that I am at the "Add Course Information" page when I provide valid information in all required fields and click the "Add Course" button, then a new course record will be successfully added to the system, and I will be redirected to the course management page. Test case: Enter 100 characters in the course name field.
- 3) Given that I am on the "Add Course Information" page, when I do not fill in all the mandatory fields and attempt to add a course by clicking the "Add Course" button, then the system will display an error message stating "Required field(s) are empty, please fill in all required information" and prompt me to complete the necessary fields.
- 4) Given that I am on the "Add Course Information" page, when I encounter a scheduling conflict with the instructor or venue while attempting to add a course, then the system will prevent the addition of the course and display an appropriate error message.
- 5) Given that I am on the "Add Course Information" page, when I click the "Cancel" button, then the system will discard any entered information and redirect me back to the course management page.

- 6) Given that I am on the "Add Course Information" page, when I enter a string with 100 characters in the course name field and a string in the power field, then the system will accept the inputs without any truncation or validation error.
- 7) Given that I am on the "Course Information Modification" page, when I update the duration of a course and click the "Save Changes" button, then the system will successfully modify the course duration, and the changes will be reflected in the system.
- 8) Given that I am on the "Course Information Modification" page, when I update the instructor of a course and click the "Save Changes" button, then the system will successfully modify the course instructor, and the changes will be reflected in the system.
- 9) Given that I am on the "Course Information Modification" page, when I update the location of a course and click the "Save Changes" button, then the system will successfully modify the course location, and the changes will be reflected in the system.
- 10) Given that I am on the "Course Information Modification" page, when I attempt to make inaccurate or conflicting changes to course information and click the "Save Changes" button, then the system will reject the changes and display an error message indicating the reason for rejection.
- 11) Given that I am on the "Course Information Modification" page, when I click the "Cancel" button, then the system will discard any changes made and redirect me back to the course management page.



Return

Text Box

Search course

Reset

1..

Edited

Delete

2 ..

Edited

Delete

3 ..

Edited

Delete

4 ..

Edited

Delete

5 ..

Edited

Delete

Add course

Plan management page

Grooming

Discussion

Related PBIs

TAS-Man-008
TAS-Man-009

Product Backlog Item

PBI #	TAS-Man-008				
Title	Course Information View and Search				
Creation Date	April 1, 2024	Status	New	Sprint	
Priority	Normal	Effort		Business Value	

Story

As the manager I want to quickly view all course information for next steps and preview all courses. Also, I want to quickly search out the courses I want to find for viewing or other actions because there are often many courses on the course page

Description

1. Need to build a page that can display all the courses through the database.
2. The page needs to be as simple and easy to view as possible.
3. There needs to be a search box to enter, a search key to click to search, and a Reset button to cancel the search.
4. The search can be any trainer, time, or location, and a matching item will appear in the search results.
5. If there are no results, display the text "no results".

Acceptance Criteria

- 1) Given that I am a valid manager, when I navigate to the "Plan Management" page, then a search box will be available for entering search queries.
- 2) Given that I am on the course page and have entered a search query in the search box, when I click the search key or press Enter, then the system will display matching courses based on the search criteria.
- 3) Given that I have conducted a search and want to reset it, when I click the "Reset" button, then the search box will be cleared, and the complete list of courses will be displayed.
- 4) Given that I am conducting a search, when the search query matches any course's trainer, time, or location, then the system will display the matching course(s) in the search results.
- 5) Given that I am conducting a search and there are no matching results, then the system will display the text "No results" to indicate that no courses match the search criteria.
- 6) Given that I am conducting a search, when I click on a matching course from the search results, then I will be directed to the detailed information page of that course.

Story

As the manager, I want to delete the information of cancelled courses to ensure the timeliness of the information and that members do not book cancelled courses.

Description

1. Courses can be deleted.
2. When deleting, a pop-up window is needed to confirm whether it is deleted.

Acceptance Criteria

1. Given that I am a valid manager, when I navigate to the "Plan Management" page, I can delete lessons.
2. Given that I am on the "Deletion of Course Information" page, when I select a course to delete and click the "Delete" button, then a confirmation pop-up window will appear to verify if I want to proceed with the deletion.
3. Given that I am presented with the confirmation popup window, when I confirm the deletion by clicking the "Yes" button, then the system will successfully delete the selected course information from the database, and the course will no longer be available for booking.
4. Given that I am presented with the confirmation popup window, when I cancel the deletion by clicking the "No" button, then the deletion process will be aborted, and the course information will remain unchanged in the system.

The screenshot displays the 'Plan management page' interface. At the top left, there is a 'Return' link with a double-left arrow icon. Below this is a search bar with a 'Text Box' placeholder, a magnifying glass icon, and two buttons: 'Search course' and 'Reset'. The main content area is a table with five rows. Each row contains a course ID (1., 2., 3., 4., 5.), a course name, and two buttons: 'Edited' and 'Delete'. A confirmation dialog box is overlaid on the table, asking 'Are you sure to delete this course?'. The dialog has two buttons: 'Confirm' and 'Cancel'. The 'Delete' button of the first row is highlighted, indicating it is the selected course for deletion. At the bottom of the table, there is an 'Add course' button. The entire interface is contained within a light blue border.

Grooming

Discussion

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Related PBIs

TAS-Man-007
TAS-Man-008

Product Backlog Item

PBI #	TAS-Mem-025
Title	Tip-in trainer

Creation Date	April 1, 2024	Status	New	Sprint	
Priority	Normal	Effort		Business Value	

Story

Suppose I am a student and I think that the trainer has a bad attitude in class or that the trainer has abused and harassed me and other unacceptable behaviors, I should have a way to tip-in the trainer to protect my rights and interests. Meanwhile, the report should not be known by the trainer, only the manager and I can see it.

Description

1. There needs to be a button on the “Personal Center” page to jump to the tip-in-trainer page after clicking it.
2. In the tip-in-trainer page, users should be able to enter the name of the trainer being tip-in, the content of the report, and their contact information.
3. There needs to be a submit button to click to submit.
4. If the content is not completed, it should be set not to submit, and you cannot click the Submit button at this time.

Acceptance Criteria

1. Since I am on the “Personal Center” page, when I click the Tip-in Trainer button, I should be redirected to the Tip-in Trainer page.
2. Within the Tip-in trainer interface, the text box filled with trainer information should be able to match the database of trainer information to ensure accuracy.
3. The tip-in content and contact information should be in the form of a text box, which should be able to enter an unlimited number of characters.
4. If everything is filled out, it should be submitted successfully.
5. If anything is not filled in, the submission should be rejected, and the user should be prompted that the filling is not complete

Grooming

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Discussion

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Related PBIs

TAS-Man-010

Product Backlog Item

PBI #	TAS-Man-010
Title	Manager view Tip-in trainer

Creation Date	April 1, 2024	Status	New	Sprint	
Priority	Insignificance	Effort		Business Value	

Story

As an manager, I should handle the relationship between members and trainers well, so when a member complains about a trainer, I should be able to see the specific information and deal with it in time

Description

1. There is a button to view tip-in that can be clicked and then redirected to a page to view tip-in.
2. On the tip-in page, I can see all the tip-in information, including the trainer who was tip-in, the reason for the tip-in and the contact information.
3. At the same time, I also need to be able to see the information of the whistleblower in order to understand the personal situation of the whistleblower.

Acceptance Criteria

1. The manager in the management page, click view tip-in, should be able to jump to the tip-in page.
2. The database of the tip-in page shares the same database as that of the member tip-in, which is written by the member and can be viewed by the manager.
3. When I click on the name of the whistleblower, I can jump to the whistleblower member's personal information interface.


View tip-in

User management

Trainer management

Plan management

Make an announcement

Renewal notice

Data report

Trainer appointment schedule

Homepage

Grooming

Discussion

Related PBIs

TAS-Mem-025